

Edge-Aware Image Fusion Techniques

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Outline

Introduction

GraphCut + Poisson + Guided Filtering

FCN + Poisson + Guided Filtering

Frequency Separation Fusion

Multi-resolution Wavelet Fusion

Luminance Transfer Fusion

Comparing Outputs

Applications

Image Fusion: Goal

- ▶ Combine complementary information from two images (e.g., *sharp structure* + *rich color*).
- ▶ Preserve structure and edges
- ▶ Avoid seams, halos and color bleeding.

GraphCut Segmentation

Objective: Isolate the foreground region to guide fusion.

Method:

- ▶ Initialize a bounding box around the target region.
- ▶ Apply GrabCut, which minimizes the energy:

$$E(L) = U(L) + V(L)$$

where:

- ▶ $U(L)$: color likelihoods for foreground/background,
- ▶ $V(L)$: smoothness enforcing spatial consistency.
- ▶ Produces a binary foreground mask selecting pixels from the sharp image.

Key Insight: Energy minimization enables iterative refinement of segmentation boundaries.

Poisson Blending

Objective: Seamlessly merge the extracted foreground into the target background.

Method:

- ▶ Use the softened GrabCut mask.
- ▶ Solve the Poisson equation:

$$\nabla^2 I = \nabla \cdot (\nabla S)$$

- ▶ Boundary values are taken from the colorful (blurred) image.
- ▶ Ensures smooth global illumination and gradient continuity.

Result: Natural-looking fused region with minimized edge artifacts.

Key Insight: Gradient-domain blending preserves smooth transitions between source and target.

Guided Filtering for Detail Reinforcement

Objective: Recover high-frequency details and maintain crisp edges.

Method:

- ▶ Apply a guided filter with local linear model:

$$q_i = a_k p_i + b_k$$

where p is the guidance (sharp image), and q is the filtered output.

- ▶ Use a gradient-based weight map to selectively enhance edges.
- ▶ Restores fine textures removed by Poisson smoothing.

Key Insight: Edge-aware refinement compensates for over-smoothing from Poisson blending.

Strengths & Limitations

Strengths

- ▶ High-quality foreground extraction with detailed edges.
- ▶ Seamless blending even under heavy background blur.
- ▶ Edge-aware refinement ensures natural transitions.

Limitations

- ▶ Requires manual bounding-box initialization.
- ▶ Segmentation degrades with low foreground–background contrast.
- ▶ Struggles with multiple simultaneous foreground objects.
- ▶ Performance depends heavily on mask accuracy.

Results

Sharp



Colorful



Init rect



GrabCut hard FG



GrabCut soft mask



Poisson result



Details sharp (vis)



Edge weight



GC+Poisson+Guided



Results

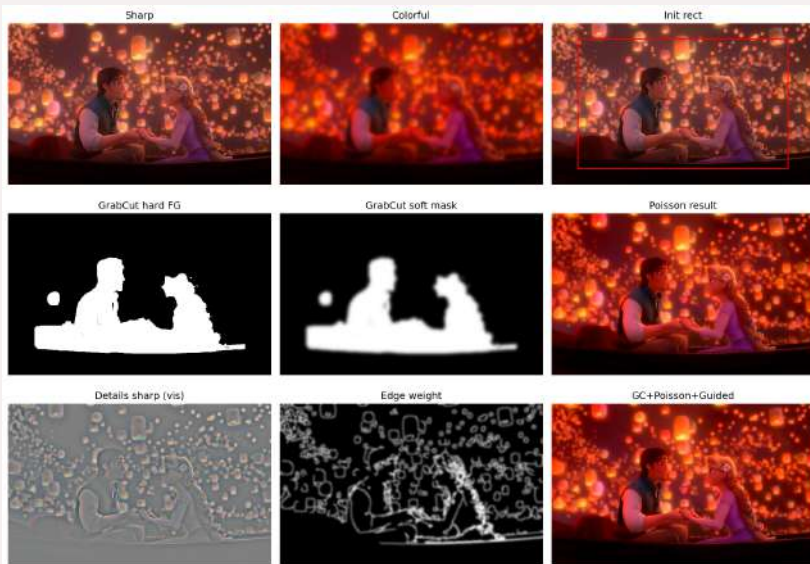


Figure: Stage wise outputs of GraphCut+ Poisson+Guided

FCN-Based Segmentation + Poisson + Guided Fusion

- ▶ The input image passes through convolution and pooling layers (encoder), building deep feature maps that capture semantic information.
- ▶ These maps are upsampled and converted into per-pixel labels using an argmax over classes.
- ▶ The resulting label map forms a foreground mask, which is softened and used to control where sharp details are injected during fusion.
- ▶ The same Poisson blending and guided filtering steps from the previous method are applied to achieve smooth merging and edge-aware detail restoration.

FCN Fusion — Strengths & Limitations

Strengths

- ▶ Automatically identifies objects using semantic segmentation.
- ▶ Works with complex and overlapping structures.
- ▶ Suitable for AR pipelines and object-aware enhancement.

Limitations

- ▶ Performance depends on FCN segmentation accuracy.
- ▶ Strongly tied to the model's training domain.
- ▶ Computationally heavier due to deep network inference.

Results

Sharp



Colorful



FCN segmentation



FCN soft mask



Poisson result



Details sharp (vis)



Edge weight



Combined weight



FCN+Poisson+Guided



Results

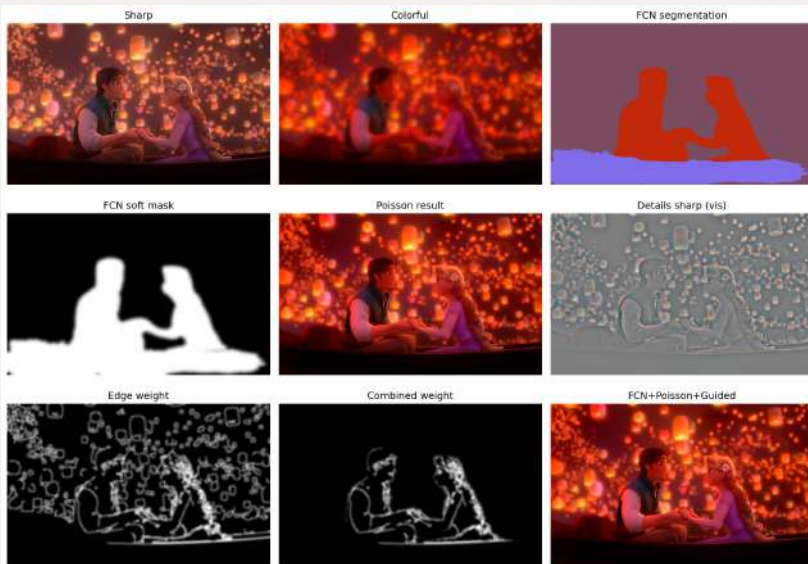


Figure: Stage wise outputs for FCN + Poisson + Guided

Frequency Separation Fusion

- ▶ **1) Low–High Frequency Decomposition:** Each image is decomposed into a low-frequency layer L (smooth color/structure) and a high-frequency layer H (edges and texture) using Gaussian filtering.
- ▶ **2) Cross-Image Frequency Selection:** Fusion is performed by taking the low-frequency components from the colorful image and the high-frequency components from the sharp image.
- ▶ **3) Edge-Preserving Reconstruction:** The fused image is reconstructed by directly summing the selected frequency layers, preserving sharp details while maintaining the color of the colorful image.

Frequency Separation — Strengths & Limitations

Strengths

- ▶ Very fast and computationally lightweight.
- ▶ Produces clear detail transfer by using high-frequency components from the sharp image.
- ▶ Maintains strong color characteristics from the colorful image.

Limitations

- ▶ Assumes a clean separation between detail and color, which may not hold in real images.
- ▶ Can cause color bleeding or loss of mid-frequency structures.
- ▶ Highly sensitive to the chosen Gaussian kernel size.

Results



Figure: Stage wise outputs for Frequency Separation Fusion

Results

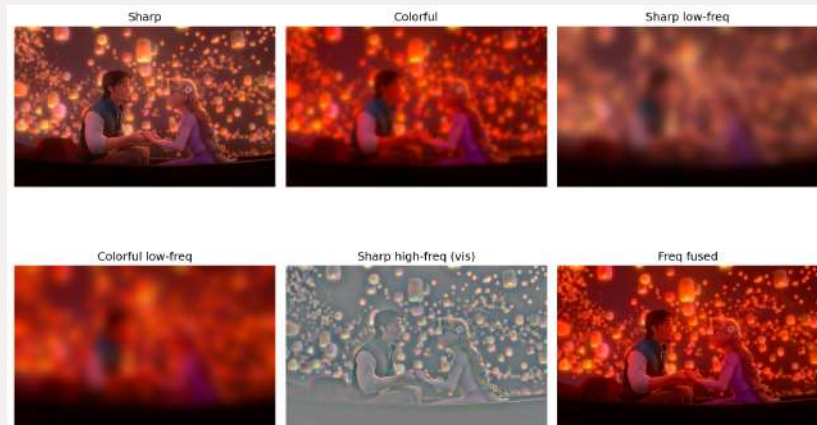


Figure: Stage wise outputs for Frequency Separation Fusion

Multi-Resolution Wavelet Fusion

- ▶ **1) Multi-Scale Decomposition:** The sharp image is decomposed into multiple detail layers using progressively larger Gaussian kernels, while the colorful image provides the coarse base layer.
- ▶ **2) Multi-Level Detail Transfer:** Fusion is performed by combining the colorful base with the rescaled detail layers taken from the sharp image.
- ▶ **3) Final Reconstruction and Enhancement:** The fused image is reconstructed by summing all components. A mild HSV saturation boost is applied to restore color richness lost during multi-scale filtering.

Wavelet Fusion — Strengths & Limitations

Strengths

- ▶ Captures details across multiple scales for balanced sharpening.
- ▶ Effective for enhancing micro-contrast without oversharpening.
- ▶ Maintains global color structure through the coarse base layer.

Limitations

- ▶ Fixed detail scaling may amplify noise or create halo artifacts.
- ▶ Highly sensitive to misalignment between the two input images.
- ▶ Multi-scale processing can reduce color saturation without post-correction.

Results

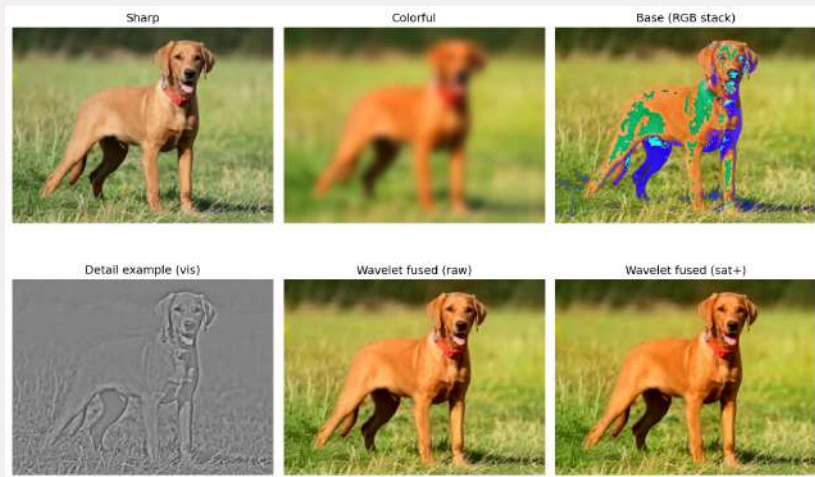


Figure: Stage wise outputs for Multi-Resolution Wavelet Fusion

Results

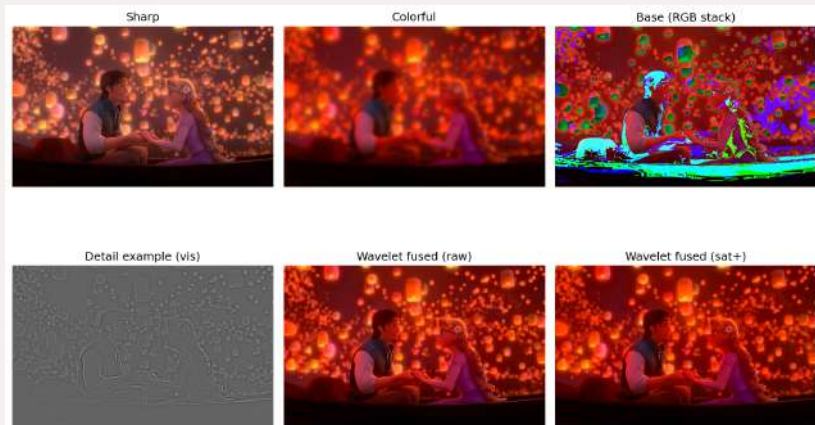


Figure: Stage wise outputs for Multi-Resolution Wavelet Fusion

Results

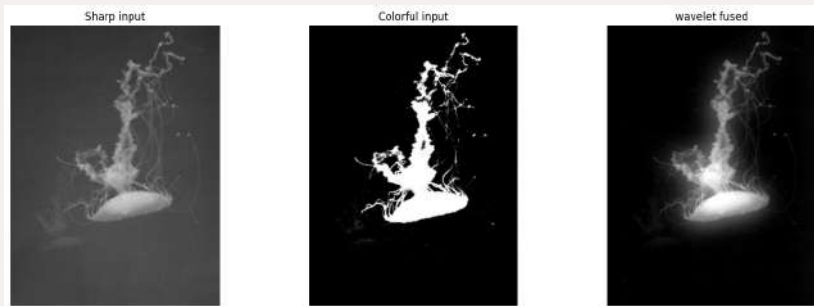


Figure: Example of wavelet producing a halo

Luminance Transfer Fusion

- ▶ **1) Conversion to CIELAB Space:** Both images are converted to CIELAB, separating luminance and chrominance:

$$I \rightarrow (L, a, b)$$

where L captures lightness and (a, b) capture color.

- ▶ **2) Luminance Replacement:** The luminance channel of the colorful image is replaced with the luminance from the sharp image, directly transferring structural brightness.
- ▶ **3) Reconstruction to RGB:** The fused LAB channels (L_f, a_f, b_f) are converted back to RGB, yielding an output with the sharp image's structure and the colorful image's chromaticity.

Luminance Transfer — Strengths & Limitations

Strengths

- ▶ Simple and efficient luminance–chrominance recombination.
- ▶ Retains strong color from the colorful image while importing structure from the sharp one.
- ▶ Effective in specialized, controlled scenarios (e.g., recoloring, illumination edits).

Limitations

- ▶ Assumes structure is only in luminance and color only in chrominance — often untrue.
- ▶ Fails on chromatic edges or features where color and structure are correlated.
- ▶ Highly sensitive to spatial misalignment between inputs.

Results



Figure: Stage wise outputs for Luminance Transfer Fusion

Results

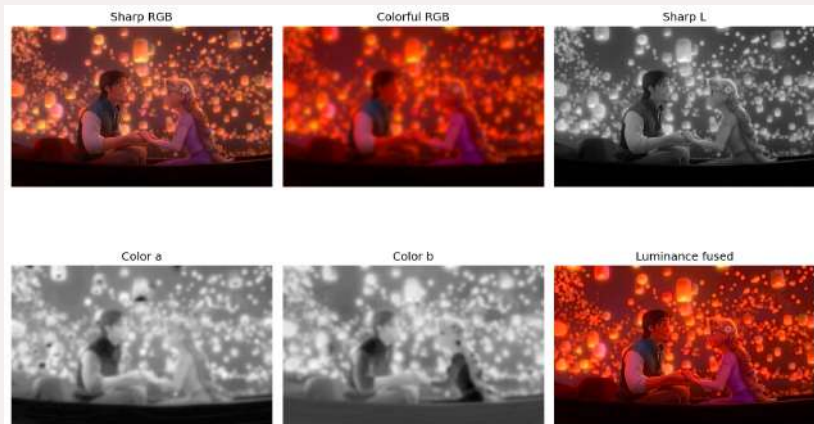


Figure: Stage wise outputs for Luminance Transfer Fusion

Results



Figure: Best case for Luminance Transfer Fusion

Results



Figure: Stage wise outputs for Luminance Transfer Fusion

Image 1 — Results

- ▶ **Sharp input:** Strong structure and fur texture.
- ▶ **Colorful input:** Warm, saturated tones but heavily blurred.
- ▶ **Frequency fused:** Restores detail but produces slight boundary harshness and uneven color mixing.
- ▶ **Wavelet fused:** Blends multi-scale texture well but slightly darkens shadows and over-enhances details.
- ▶ **Luminance fused:** Maintains natural contrast but introduces mild color shifts in highlights.
- ▶ **FCN + Poisson + Guided:** Good structure and color integration but slightly over-smooths uncertain background regions.
- ▶ **GraphCut + Poisson + Guided:** Cleanest result with sharp fur, natural saturation, and smooth background blending.

Sharp input



Colorful input



Freq fused



Wavelet fused



Luminance fused



FCN+Poisson+Guided



GC+Poisson+Guided



Figure: Image 1

Image 2 — Results

- ▶ **Sharp input:** High structural detail but lacking the warm colors of the colorful image.
- ▶ **Colorful input:** Rich saturation and glow but heavily blurred with weak structure.
- ▶ **Frequency fused:** Restores sharpness but introduces halo artifacts and inconsistent color mixing.
- ▶ **Wavelet fused:** Improves multi-scale detail but slightly darkens faces and reduces color consistency.
- ▶ **Luminance fused:** Strong contrast but suffers from unnatural color shifts due to luminance–chroma mismatch.
- ▶ **FCN + Poisson + Guided:** Object-aware and sharp but slightly over-saturated where segmentation is uncertain.
- ▶ **GraphCut + Poisson + Guided:** Most natural color–detail balance with clean boundaries thanks to user-guided masking.



Figure: Image 2

Applications

- ▶ **GraphCut + Poisson + Guided:** Best for interactive editing, clean foreground extraction, portraits, and background replacement.
- ▶ **FCN + Poisson + Guided:** Ideal for AR, automatic object enhancement, multi-object scenes, and exposure-aware fusion.
- ▶ **Frequency Separation Fusion:** Great for real-time stylization, fast color–detail merging, and lightweight mobile filters.
- ▶ **Wavelet Fusion:** Suited for photography-style enhancement, micro-contrast improvement, and multi-scale sharpening.
- ▶ **Luminance Transfer Fusion:** Useful for recoloring, illumination changes, style-transfer-like effects, and perfectly aligned images.